“SAMVED” HACKATHON 2026

सोलापूर महानगरपालिका, सोलापूर

MIT Vishwaprayag University

TITLE PAGE

- **Problem Statement ID –** **PS004**
- **Problem Statement Title –** “Smart Safety and Assistance System for Sanitation Workers of Solapur Municipal Corporation”
- **Theme –** Public Safety & Social Welfare
- **Team ID –** **302B28A4**
- **Team Name (Registered on portal) –** **RAKSHAK TECH**

Real-Time Sewage Pit Hazard Monitoring & Worker Safety System

RAKSHAK TECH

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Addressing the Problem

- Sanitation workers enter sewer pits without proper hazard checks and face toxic gas and suffocation risks.
- There is no live monitoring of underground conditions or worker health during operations.
- Emergency alerts depend on manual communication, which often fails underground.

Uniqueness & Innovation

- **Pre-Entry Smart Probe :** Hazard scan before entry
- **Underground P2P Network :** Works without internet underground
- **Entry Decision Engine :** Auto safe/unsafe entry approval
- **Worker + Pit Linked Monitoring :** Combined environment and vitals monitoring

NOTE :- Please Find the detailed problem overview and the detailed proposed solution here

Proposed Solution

Sensor data

Links pit stations, probes & wearables for continuous underground safety monitoring

Wearable Safety Devices

- Pre-entry hazard scan + live pit monitoring
- Works even without network
- Worker vitals + fall + distress alerts
- Pit risk history & citywide safety analytics

Server & Mobile App

Stakeholders

Decision Support

Enables risk trend detection, hazard prediction, and safer entry planning for all safety teams.

"One Safety Network, Many Protections"

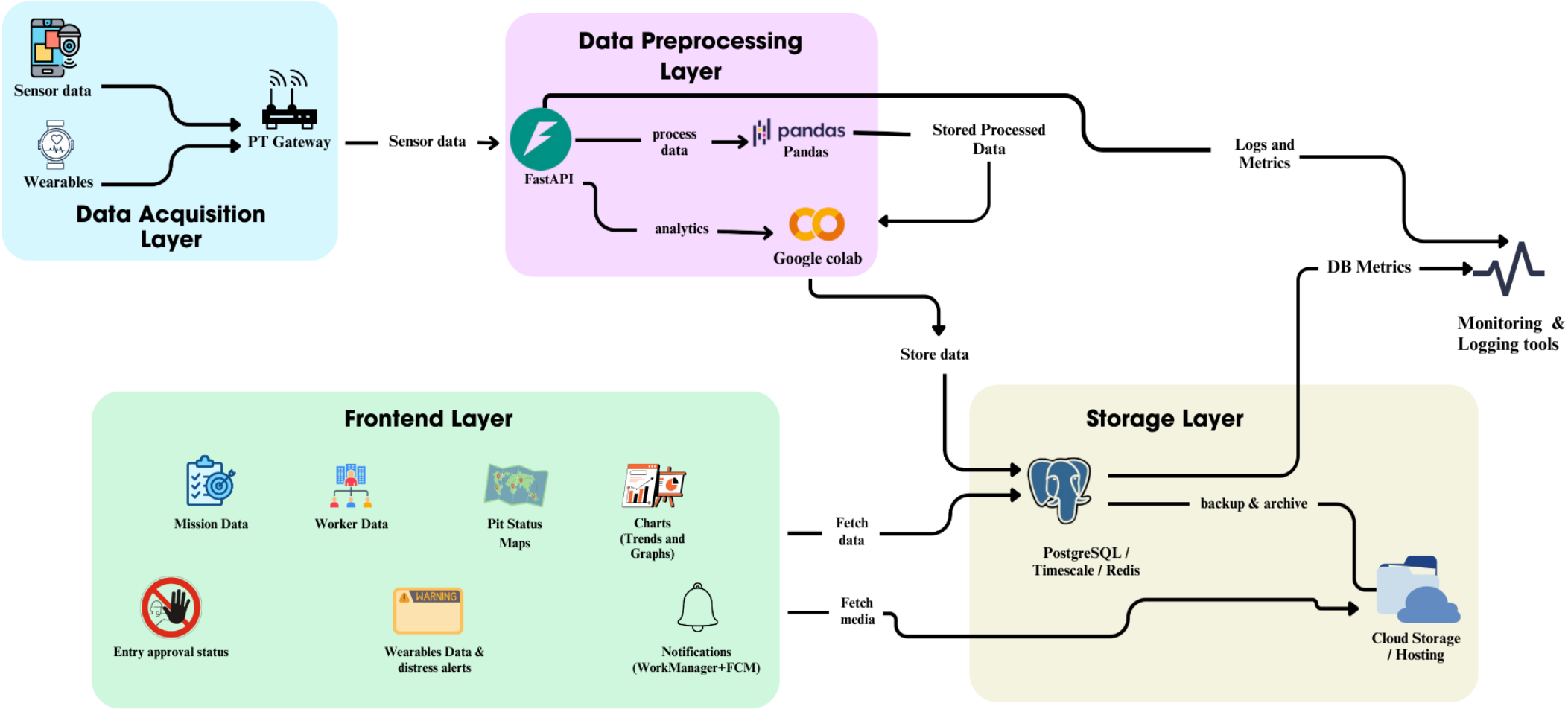
- Workers: real-time hazard & distress alerts
- Supervisors: live pit status & entry approval
- Municipality: risk heatmaps & planning data
- Safety teams: faster emergency response

TECHNICAL APPROACH

TECH STACK

- Flutter
- FastAPI
- Redis
- PostgreSQL
- Pandas
- Google colab
- Android studio
- Arduino IDE
- LiprePCB
- VS Code

SYSTEM ARCHITECTURE



FEASIBILITY AND VIABILITY

Feasibility

- Internationally trusted & tested sensors & Tech used
- Open-source Tech stack
- Quick Prototype → Field deploy
- Realtime notifications + visualizations

Viability

- Tackles Hazardous diseases before coming in contact
- Converts Sensor Data → actionable insights
- "One Data, Many Stories" (multi-stakeholder system)
- Low-cost, scalable solution

Challenges

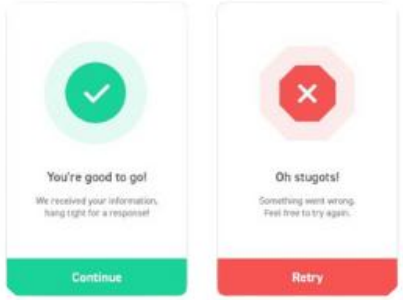
- Underground Network Coverage
- Sensor drift / Calibration needed
- Non technical user adoption
- Scaling to multiple locations at same time

Strategies

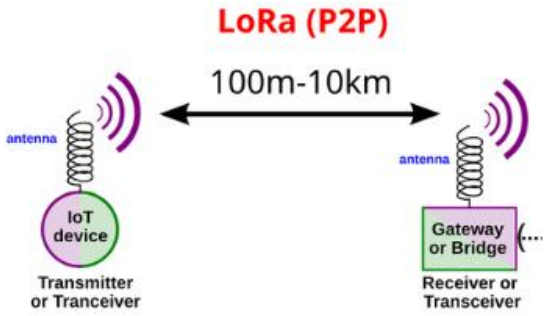
- LoRa P2P communication [No network required]
- Automatic sensor confidence checks
- Icon-based simple UI for workers & non-technical users
- Cloud ready backend for future growth

IMPACT AND BENEFITS

Real-time hazard detection prevents toxic exposure and sewer fatalities
(Worker Safety First)



Smart IOT-assisted gas detection probe entry approval replaces blind manual sewer entry



Always-Connected Underground

Peer-to-peer LoRa safety network works even without internet

Multi-pit data builds hazard maps and preventive action planning

Prevention over Rescue

Detects danger before worker entry, not after incidents

Converts gas + vital sensor's Data into clear Safe / Warning / Danger signals

Low-Cost, High-Impact

Affordable hardware with scalable smart-city deployment

Sensors **Self-diagnosing** and **Auto-calibrating** themselves reduces error risk and improves safety confidence

RESEARCH AND REFERENCES

RESEARCH

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